

Automated Referee Assignment Engine

Optimized via Multi-Constraint Greedy Selection Algorithm

Ronak Shah, 28 June 2026, 617.680.9439, djkanor@gmail.com

Problem

- Referee assignments are done manually through the AdminSports portal

This makes tracking important referee metrics difficult

- We have found issues around equal game distribution challenging to quantify

Proposed solution

Enable referee assignors to use a Google Spreadsheet driver input file to provide per-referee time windows and generate a schedule that is delivered seamlessly to AdminSports backend via API and generates assignment emails and calendar links

Ensure that referee assignments account for age and experience constraints as a ruleset —> potentially also captured via a Google Spreadsheet driver file but processed by natural language rules

Produce a visual metrics table that illustrates

- years of refereeing (decimal)
- number of assigned games
- completed report status
- number by grade-age group
- number by gender
- yellow and red card distributions to players and coaches

Current implementation

Python script consumes:

- an example game schedule from Parkway United
- a referee list with fake birthdates, start and stop times

Rule set for assigning referees is only age based right now (ran out of GitHub CoPilot credit)

Rule set is not modular

Desired rule set

Rule 0: (**All centers first**): ensure that all centers are assigned first, before assigning ARs

Rule 1 (**High-Stakes Centers**): Grades 6 & 8 (Divisions 1 & 2) require a Center Referee who is ≥ 18 years old OR has ≥ 2 years of experience

Rule 2 (**High-Stakes ARs**): Assistant Referees (ARs) for Grade 8 (Divisions 1 & 2) must have ≥ 0.5 years of experience

Rule 3 (**Junior Development**): Grades 3 & 4 Center Referees can be any age, but the algorithm preferentially selects candidates < 18 years old

Rule 4 (**Mid-Grade Centers**): Grade 5/All divisions & Grade 6/Division 3+ Center Referees must have ≥ 0.5 years of experience

Rule 5 (**Open Slots**): ARs for Grade 5 (all divisions) and Grades 6 & 8 (Divisions 3+) have no age or experience restrictions

Rule 6 (**Load management**): referees can do no more than 3 consecutive games, and only 1 at center

Referee gap identification

The greedy rule set assumes sufficient numbers of referees

For any given ruleset, this condition may not hold

Future algorithms must identify gap situations and alert the referee assignor when the algorithm is not able to complete assignments

Ways forward to address gaps:

- Assign all game centers and override age constraints for Grade 6 and younger regardless of divisions
- Identify games WITHOUT ARs for Referee assignor reviews in AdminSports portal (include email to assignor as trigger)
- Generate alert messages — after referee assignor **approval** of the truncated scheduling — to home and away coaches for their situational awareness

Additional TODOs

1. (1) Test Google spreadsheet lookup that integrates with Google workspace documentation for club's referee listing. This is where we plan to centralize and secure referee information for week-to-week planning
2. Build and test API connectivity to AdminSports
3. Implement the desired logic as a separate look-up to ensure modularity

Metrics Tracker #1: Referee performance metrics summary

Referee Performance & Statistics Tracking							
Individual Referee Metrics - Seasonal Data							Export to CSV
Referee Name	Years Experience	Number of Assignments	Number of Reported Games	Number of Centers	Number of ARs	Percentage of Games Centered	Percentage of Games AR
David Miller	8	35	39	28	13	80%	37%
Sarah Jones	6	45	43	19	26	42%	58%
Michael Brown	11	55	52	38	17	57%	69%
Jaran Loi	3	29	27	8	21	58%	60%
Elizabeth Smith	8	42	39	28	14	39%	40%
Chris Patel	12	48	43	19	17	42%	43%
Robert Davies	2	18	15	9	9	15%	17%
Laura Kim	3	29	27	8	7	12%	17%
William Green	8	36	42	17	14	42%	58%
Katherine Lee	6	45	43	19	26	43%	46%
Thomas Moore	4	20	25	8	10	10%	10%
Jessica Brown	2	29	27	8	21	20%	21%

Metrics Tracker #2: At a glance report of accepted and submitted reports

Referee Assignment & Game Tracking

REFEREE PROGRESS TRACKER - UPCOMING & COMPLETED GAMES

GAME DATE & TIME	HOME TEAM	AWAY TEAM	CENTER REFEREE		ASSISTANT REFEREE #1		ASSISTANT REFEREE #2	
				Game Reporting	Acceptance	Game Reporting	Acceptance	Game Reporting
Saturday, Oct 26 3:00 PM EST	Middletown United	Oakwood Strikers	David Miller					
Saturday, Oct 26 3:00 PM EST	Bayfield FC	Riverdale City	Accepted	A A A	Sarah Jones	A A	Michael Brown	A
Saturday, Oct 27 3:00 PM EST	Bayfield FC	Riverdale City	Accepted	A A A	Sarah Jones	A A	Sarah Jones	A
Saturday, Oct 28 3:00 PM EST	Bayfield FC	Oakwood Strikers	Accepted	A A A	Sarah Jones	A A	Sarah Jones	A
Saturday, Oct 27 3:00 PM EST	Bayfield FC	Riverdale City	Accepted	A A A	Sarah Jones	A A	Michael Brown	A
Saturday, Oct 28 3:00 PM EST	Middletown United	Oakwood Strikers	Accepted	A A A	Sarah Jones	A A	Michael Brown	A
Saturday, Oct 29 3:00 PM EST	Middletown United	Riverdale City	Accepted	A A A	Sarah Jones	A A G	Jamh Lol	A A
Saturday, Oct 20 3:00 PM EST	Riverdale City	Bayfield FC	Accepted	A A A	Sarah Jones	A A G	Sarah Jones	A A
Saturday, Oct 30 3:00 PM EST	Bayfield FC	Oakwood Strikers	Accepted	A A A	Sarah Jones	A A	Michael Brown	A G
Saturday, Oct 30 3:00 PM EST	Riverdale City	Bayfield FC	Declined	A A A	Sarah Jones	A A G	Jaran Lol	A G